

House Price Prediction Using Linear Regression

Internship Project Report

Submitted By

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Domain

Machine Learning Internship

Project Title

House Price Prediction Using Linear Regression

1. Objective

The objective of this project is to predict house prices using machine learning techniques. The model uses various house features such as bedrooms, bathrooms, living area, floors, and grade to estimate the market price of a house.

2. Tools and Technologies Used

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn
- Joblib

3. Dataset Description

The dataset contains information about residential properties and their selling prices.

Dataset Information

- Dataset Name: House Sales Prediction Dataset

- Source: Kaggle
- Target Variable: Price

Important Features

- Bedrooms
- Bathrooms
- Square Feet Living Area
- Floors
- Waterfront
- View
- Grade

Target Variable

- Price

4. Exploratory Data Analysis (EDA)

EDA was performed to understand the structure and quality of the dataset.

Operations Performed

- Displayed first five records using `head()`
- Checked dataset information using `info()`
- Generated statistical summary using `describe()`
- Checked missing values using `isnull().sum()`

Observations

- Dataset contains both numerical and categorical information.
- Price is the target variable.
- No significant data quality issues were observed.
- Several features show strong relationships with house prices.

Screenshots

df.head()

The screenshot shows a JupyterLab environment with a file named 'house_price_prediction'. The last checkpoint was 13 minutes ago. The interface includes a menu bar (File, Edit, View, Run, Kernel, Settings, Help) and a toolbar with icons for file operations and code execution. The Python 3 (ipykernel) environment is selected.

The code cell [1] contains the following imports:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

The code cell [2] contains the following code:

```
df = pd.read_csv("kc_house_data.csv")
df.head()
```

The output of the code cell [2] shows the first 5 rows of the dataset:

	id	date	price	bedrooms	bathrooms	sqft_living	sqft_lot	floors	waterfront	view	...	grade	sqft_above	sqft_basement	yr_built	yr_re
0	7129300520	20141013T000000	221900.0	3	1.00	1180	5650	1.0	0	0	...	7	1180	0	1955	
1	6414100192	20141209T000000	538000.0	3	2.25	2570	7242	2.0	0	0	...	7	2170	400	1951	
2	5631500400	20150225T000000	180000.0	2	1.00	770	10000	1.0	0	0	...	6	770	0	1933	
3	2487200875	20141209T000000	604000.0	4	3.00	1960	5000	1.0	0	0	...	7	1050	910	1965	
4	1954400510	20150218T000000	510000.0	3	2.00	1680	8080	1.0	0	0	...	8	1680	0	1987	

5 rows x 21 columns

df.info()

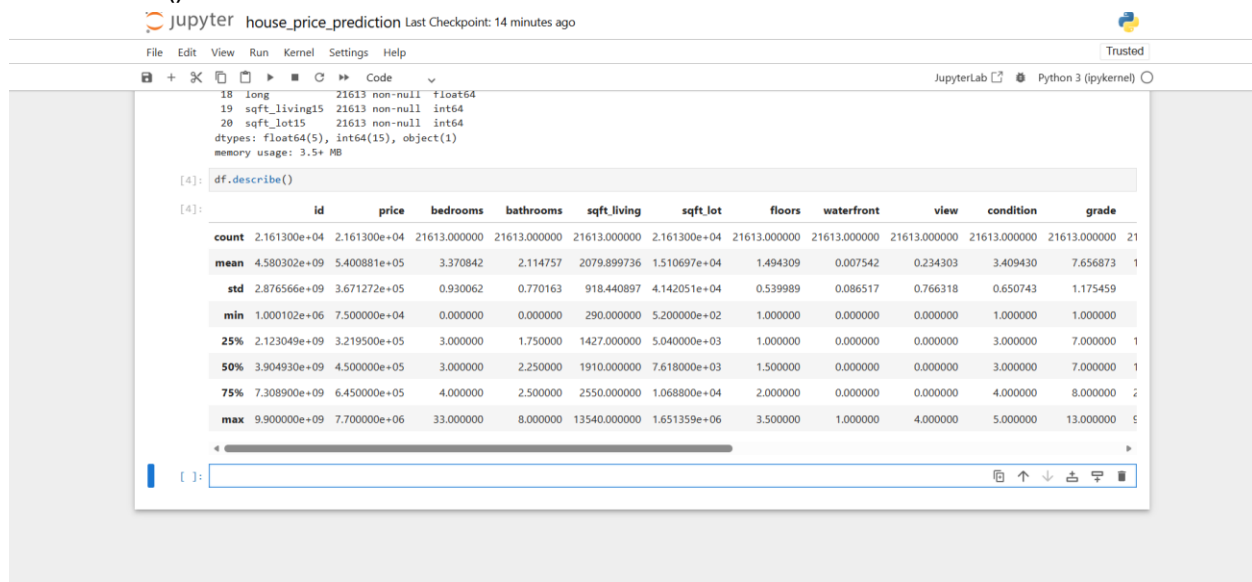
The screenshot shows the same JupyterLab environment. The code cell [3] contains the following code:

```
df.info()
```

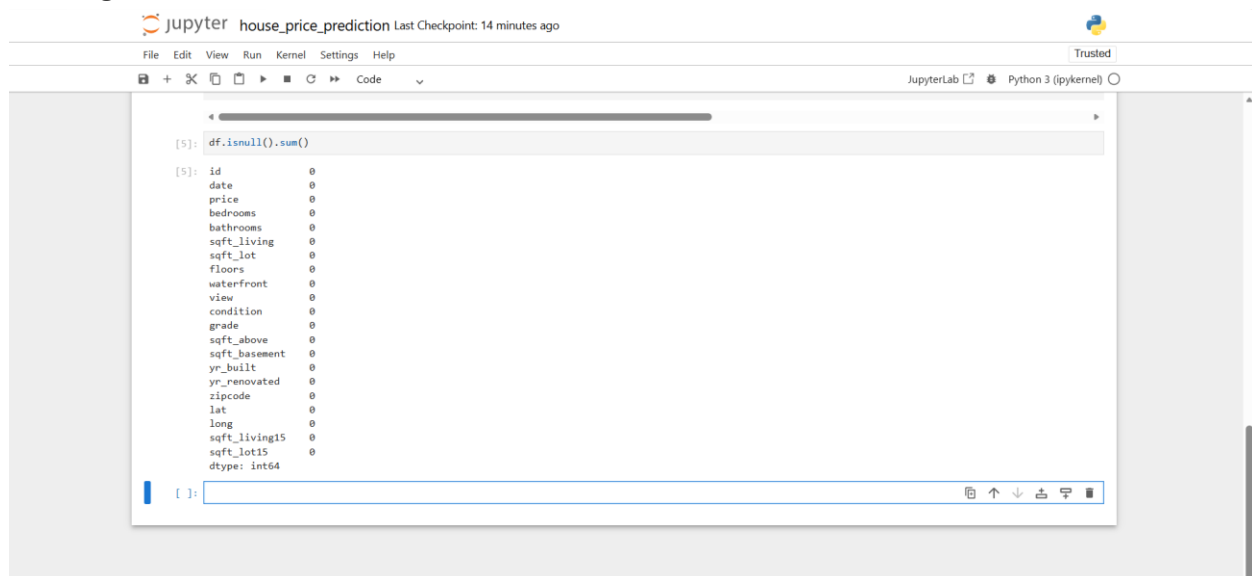
The output of the code cell [3] shows the information about the DataFrame:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 21613 entries, 0 to 21612
Data columns (total 21 columns):
 #   Column              Non-Null Count  Dtype  
---  --
 0   id                   21613 non-null  int64  
 1   date                 21613 non-null  object  
 2   price                21613 non-null  float64 
 3   bedrooms             21613 non-null  int64  
 4   bathrooms            21613 non-null  float64 
 5   sqft_living          21613 non-null  int64  
 6   sqft_lot             21613 non-null  int64  
 7   floors               21613 non-null  float64 
 8   waterfront           21613 non-null  int64  
 9   view                 21613 non-null  int64  
10   condition            21613 non-null  int64  
11   grade                21613 non-null  int64  
12   sqft_above           21613 non-null  int64  
13   sqft_basement        21613 non-null  int64  
14   yr_built             21613 non-null  int64  
15   yr_renovated         21613 non-null  int64  
16   zipcode              21613 non-null  int64  
17   lat                  21613 non-null  float64 
18   long                 21613 non-null  float64 
19   sqft_living15        21613 non-null  int64  
20   sqft_lot15           21613 non-null  int64  
dtypes: float64(5), int64(15), object(1)
memory usage: 3.5+ MB
```

df.describe()



Missing Values



5. Data Visualization

Four visualizations were created to understand patterns within the dataset.

Visualizations Performed

1. House Price Distribution Histogram
2. Living Area vs Price Scatter Plot
3. Price Box Plot

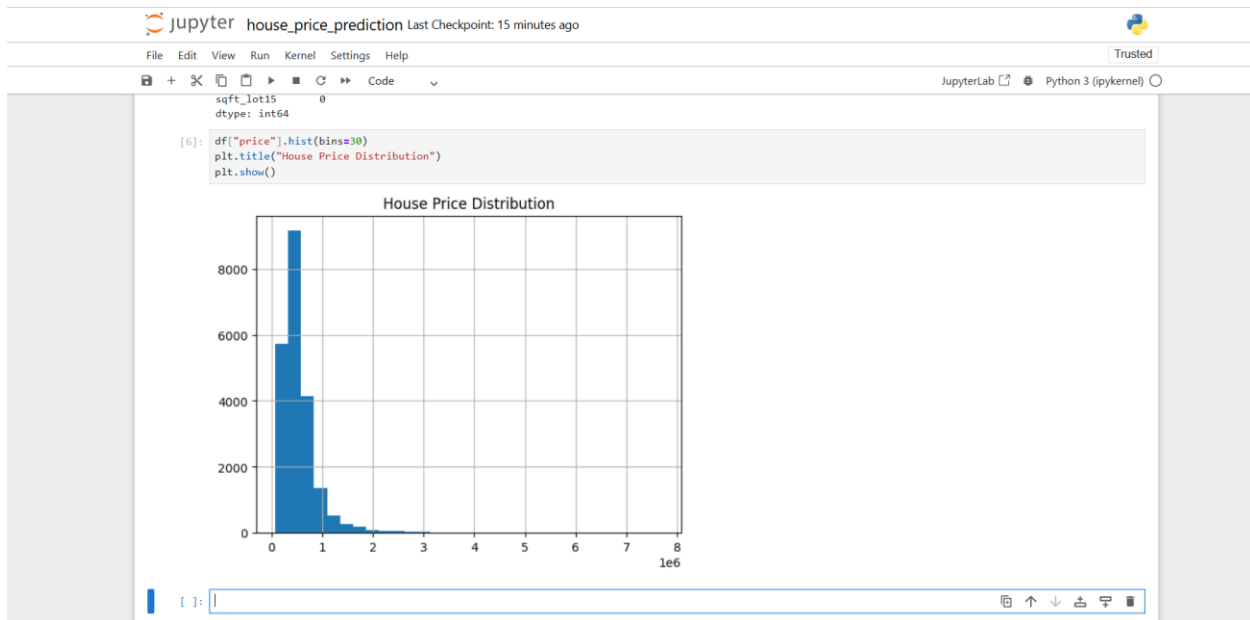
4. Correlation Heatmap

Observations

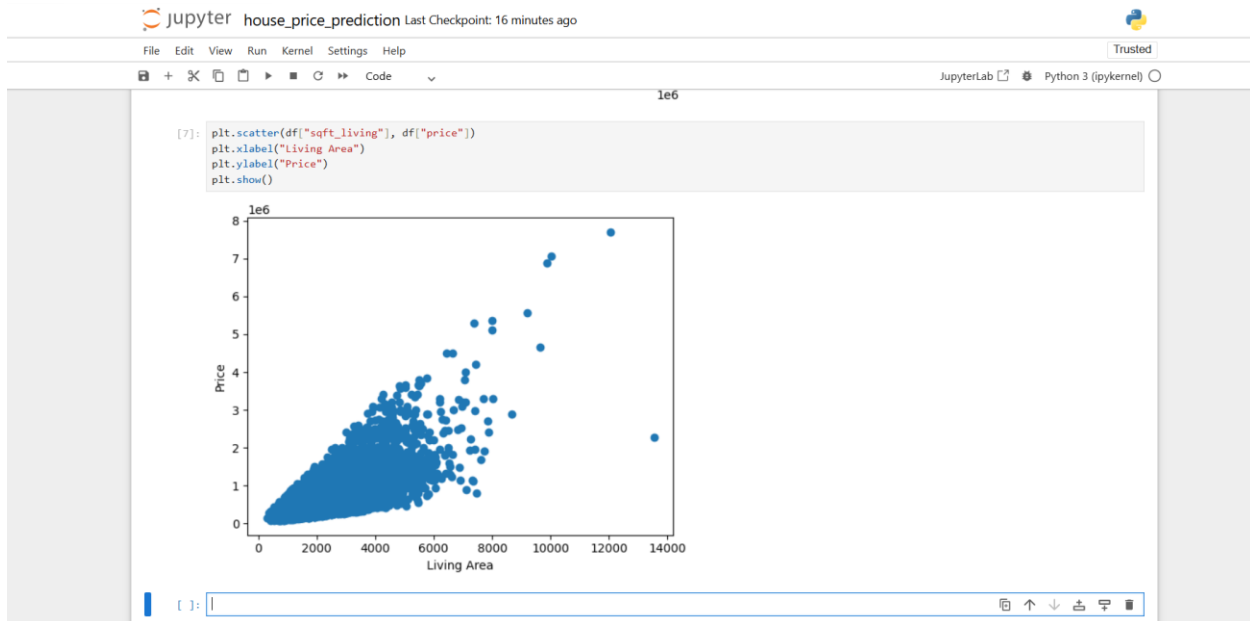
- House prices are not uniformly distributed.
- Houses with larger living areas generally have higher prices.
- Some price outliers are present.
- Features such as living area and grade show positive correlation with price.

Screenshots

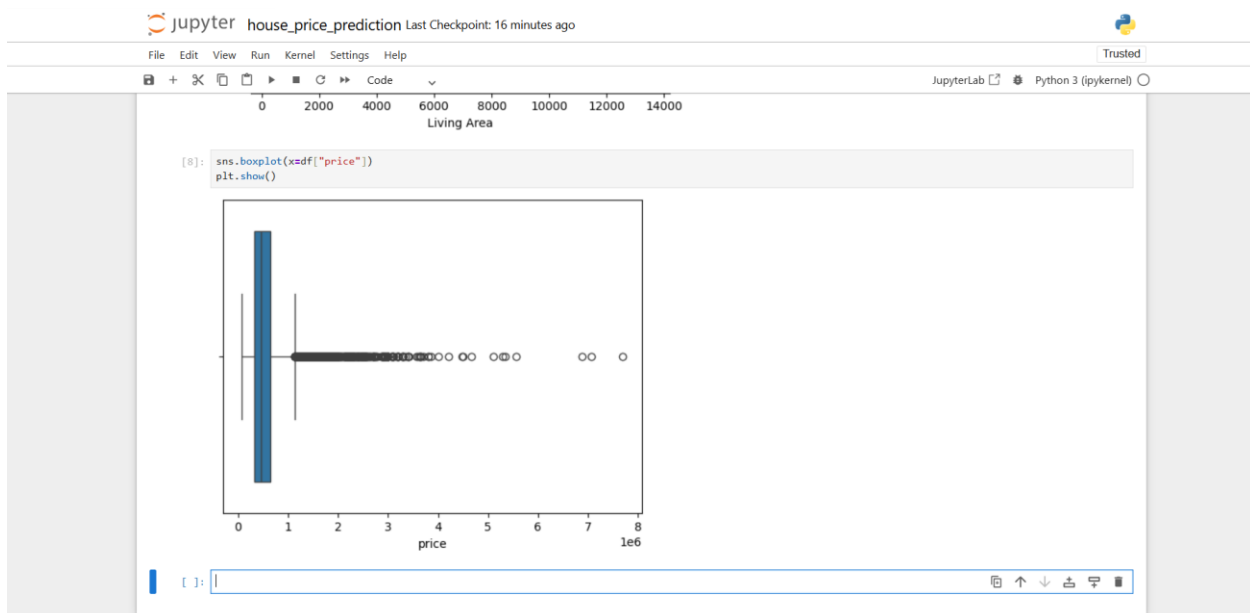
Histogram



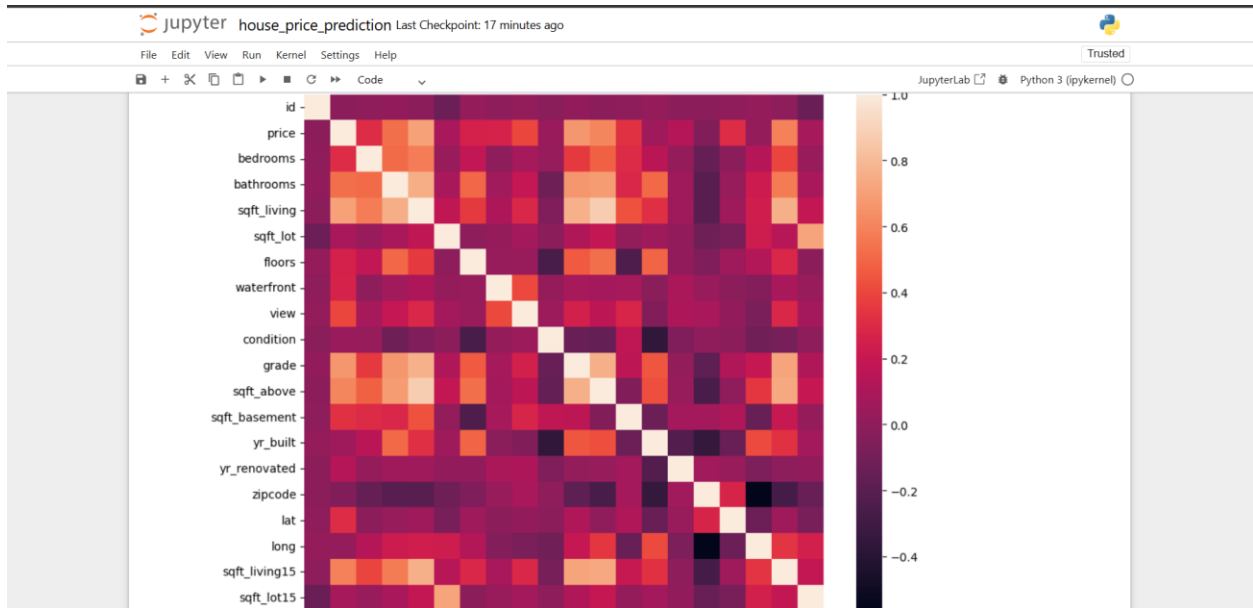
Scatter Plot



Box Plot



Correlation Heatmap



6. Feature Selection

The following features were selected for model training:

- Bedrooms
- Bathrooms
- Square Feet Living Area
- Floors
- Waterfront
- View
- Grade

Target Variable:

- Price

7. Model Building

Algorithm Used

Linear Regression

Steps Performed

1. Feature Selection

2. Train-Test Split
3. Model Training
4. Prediction on Test Data
5. Model Evaluation

8. Model Evaluation

Evaluation Metrics

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- R^2 Score

Results

MAE: 157757.23555980396

MSE: 60027177022.57606

R^2 Score: 0.6029337050233837

9. Model Saving

The trained model was saved using Joblib.

File Name:

house_price_model.pkl

The saved model can be loaded later and used for future house price predictions without retraining.

10. Conclusion

The House Price Prediction project was successfully completed using Linear Regression. Data preprocessing, exploratory data analysis, visualization, feature selection, model training, and evaluation were performed successfully.

The model demonstrated the ability to estimate house prices based on important property features. This project provided practical experience in regression analysis and predictive modeling using machine learning.

11. Repository Artifacts

Files Included:

- house_price_prediction.ipynb
- house_price_model.pkl
- house_report.pdf
- Dataset File
- Screenshots

Submitted as part of the InternSpark Machine Learning Internship Program.